

marijuana, open legal marijuana dispensaries, and interactions between an indicator that the state had a triplicate prescription program and year fixed effects.²⁸ We continue to find a large and statistically significant decrease in mortality with the inclusion of these controls.

Next we examine the sensitivity of our results to a control for different exposure to trade from China (i.e., the “China Shock” described in Autor et al., 2013, 2019). This previous research has connected increased trade competition with declining economic and social conditions, which may in turn affect health. We allow counties to have different time trends by interacting the commuting-zone level of exposure to Chinese imports per worker from 2000 to 2014 with year fixed effects.²⁹ The inclusion of this control variable does not impact our mortality estimate.

Finally, in row 9, we control for the demographic composition of our sample by adding individual covariates for race, age, and gender to our model. Once again the results are largely unchanged by the inclusion of these covariates. We also estimate a model (in row 10) that includes all of the controls described above. We continue to find a negative and statistically significant impact of the ACA Medicaid expansions on mortality rates in our sample, and cannot reject that the effect estimated in this model is significantly different than our main result.

One separate concern is that a small subset of our sample ages into Medicare during the 2015-2017 years in the post-period. If the Medicare program differs systematically across states in a manner that is correlated with Medicaid expansion, we might inadvertently be picking up differential mortality effects under the Medicare program. To rule out this concern, we re-estimate our model for the subgroup ages 55-61 in 2014, who do not qualify for Medicare during the post period. The results from this analysis may be found in Appendix Table A7 and are similar to those for our main sample.

VI.D Placebo Tests

To further assess the validity of our empirical approach, we conduct several “placebo” tests. In these tests, we investigate whether we observe effects of the Medicaid expansions in populations that were unaffected or less affected by the policy change.

Our first placebo test shifts the analysis sample back in time to the pre-ACA period. This test can assess whether any elements of our sample construction, such as drawing the ACS sample only in the pre-expansion period, might lead to spurious results. We construct the data in the same fashion as our main analysis, but use mortality data from 2004 to 2013 for ACS respondents in the 2004 to 2009 survey years (rather than mortality data from 2008 to 2017 for the 2008 to 2013 survey years). We construct a variable indicating that a state expanded that corresponds to $Expansion_s$ in equation (1), but estimate our model as if the first expansions occurred in 2010 rather than 2014, with states expanding t years after 2014 treated as if they expanded in 2010+ t . The results of this placebo test using the pre-ACA period is presented in the first row of Figure IV. As expected, we find no effects on Medicaid coverage or mortality in expansion states during this placebo pre-ACA period.

Next, we expand on this analysis by randomly assigning Medicaid expansion status to the same number of states and years as occurred under the ACA Medicaid expansions, rather than estimating

²⁸We draw heavily on Alpert et al. (2019) to identify these opioid related policies, following the sources and coding outlined in their paper. See Appendix Section 4 for additional details.

²⁹Note that these measures are not available for Alaska and Hawaii, so we drop these two states when estimating this specification.

placebo effects for the actual expansion states. As with the previous exercise, we conduct this placebo test using pre-ACA data, prior to when the actual expansions occurred. We repeat this exercise 10,000 times and compile the coefficient estimates and t-statistics from the difference-in-differences model. The results are presented in Appendix Figure A4, with the vertical lines indicating the 5th and 95th percentiles of this “placebo” distribution; our model estimate is depicted with a dashed line. This exercise allows us to further investigate the likelihood that we might encounter similarly sized mortality effects by chance. Our estimates for the ACA Medicaid expansions fall well below the 5th percentiles of the distributions of placebo coefficient estimates and t-statistics, further increasing our confidence that we are estimating a true policy effect.

We next examine whether we observed similar mortality effects for two population groups that were less likely to be affected by the policy. The first test uses individuals who were age 65 and older at the time of the expansions. These individuals have near universal coverage through the Medicare program and should not be directly affected by the coverage expansions.³⁰ We re-estimate equation (1) for this sample and the results are presented in the second panel of Figure IV. As predicted, we observe no effect of the Medicaid expansions on Medicaid coverage for this group. We also see no effect of the expansions on mortality rates for this group.³¹

We next examine individuals ages 55 to 64 in families earning 400% FPL or greater at the time of the ACS survey. This group should be less affected than our main sample of low-income or low-education respondents. However, they may still gain Medicaid coverage under the expansions due to changes in income over time, or if their income is reported with error. As seen in the third row of Figure IV, we do find small but statistically significant increases in Medicaid enrollment associated with the expansions among this group. We also see correspondingly small but, for some years, statistically significant reductions in mortality for this group. These changes are consistent with a causal effect of expanded Medicaid coverage on mortality. One interesting note is those reporting higher incomes in the survey, but who later enroll in Medicaid, exhibit very high average mortality compared to those who do not enroll in Medicaid. This indicates that individuals in the high income group who do enroll in Medicaid are highly selected (e.g. having experienced a serious health event that resulted in income loss).³²

We also formally test for a differential effect in our main sample relative to the age 65+ and 400% FPL+ placebo groups by estimating a “triple difference” model. Details on this approach are found in Appendix Section 6. The “triple difference” estimates reported in the bottom two rows of Figure III are slightly smaller than our main DD results but remain statistically and economically significant.

VII Additional Analyses

We also conduct several additional analyses on different samples and subgroups to further characterize the impact of the Medicaid expansions. First, we examine changes in mortality for all nonelderly adults

³⁰Prior work has documented some spillover effects on the health care utilization of this population under pre-ACA state Medicaid expansions, but analyses of the ACA Medicaid expansions find no evidence of such spillovers and are able to rule out very small effects (Carey et al., 2018).

³¹Results are similar if we also restrict the elderly to be in low-income families.

³²If we scale the average mortality effect by the corresponding estimate for the change in any Medicaid coverage, the decrease for the high-income group is larger in size than for our main sample. We believe this reflects a larger role of adverse selection for the higher-income group.

and other age subgroups. Our main analysis is limited to individuals ages 55 to 64 at the time of the Medicaid expansions, a group with higher mortality rates that has been the focus of other work on this topic (e.g. [Black et al., 2019](#)). In the first row of Appendix Table A9, we also estimate the impact of Medicaid expansion on mortality for individuals who meet our sample inclusion criteria but are ages 19 to 64 in 2014. We find a significant increase in enrollment in any Medicaid coverage of 12.7 percentage points. The mortality estimate is not statistically significant but suggests a 3.9 percent decrease relative to the counterfactual rate. We next estimate the effects for other subgroups in this age range (ages 19-29, 30-39, 40-49, and 50-54). We find significant increases in Medicaid enrollment for all groups, but less evidence of corresponding mortality changes. The largest decrease in mortality is observed at ages 50-54 but is not statistically significant. We are unable, however, to rule out meaningful declines in mortality for any age group.

Next, we further probe heterogeneity by age within our main age group of interest, adults age 55 to 64 years old in 2014, and at older ages. In Appendix Figure A5 we report the percent reduction in mortality associated with the Medicaid expansions in 3-year age bins. We construct this by dividing the reduced form difference-in-differences estimate by the counterfactual mortality rate for each age group. We see that the largest mortality reductions occur at the oldest age groups, with those age 59-61 and 62-64 both experiencing statistically significant reductions in mortality. Reductions in mortality are smaller and not statistically significant for those at younger ages (53-55 and 56-58) and for those age 65 and older in 2014, who were Medicare eligible.

We next examine the effects of the expansions on different subgroups of our main analysis sample defined by race and ethnicity, gender, marital status, or other characteristics. The results of these analyses are reported in Appendix Table A10. We find evidence of larger mortality effects for white, non-Hispanic adults when compared to other racial and ethnic groups. Surprisingly, we detect no effect of the expansions on non-Hispanic Black respondents. We also find larger effects for the males in the sample. These differences are present despite sometimes smaller first stage estimates for these two groups, although their estimated mortality risk is higher (as seen by a comparison of their counterfactual mortality rates). In addition, the confidence intervals on the mortality estimates do not rule out meaningful declines in mortality for any subgroup. We did not find evidence of mortality differences by marital status. And, when we narrow the socioeconomic criteria to either less than a high school degree or less than 138 percent of the FPL, we find overall very similar first stage and mortality estimates for the two groups. Within this set of heterogeneity analyses, there does not appear to be a strong correlation between the size of the first stage and the effect of the expansion on mortality. However, we note that different groups tend to experience mortality from different causes, some of which may be more or less amenable to health interventions. Across subgroups there may also be differential barriers to seeking and receiving care. We leave further exploration of this heterogeneity for future work.

Finally, an additional analysis limits the main sample to approximately a 30 percent subset who reported being uninsured at the time of the survey. As this group is slightly younger on average,³³ the counterfactual mortality rate is a bit lower than in the overall sample, at 1.46 percent per year (vs. 1.63 percent in the sample overall). The point estimates indicate somewhat larger decreases in mortality

³³We estimate that the average age of uninsured respondents in our sample is 59.0 in 2014, compared to an average of 59.4 years for insured respondents. This difference is statistically significant with a p-value < 0.0001.

for this group of 0.150 percentage points (or 10.3 percent of the counterfactual rate) compared to the reduction in the main sample of 0.132 percentage points (or 8.1 percent of the counterfactual rate). The fact that those who were uninsured prior to the ACA experienced larger mortality improvements as a result of the policy is in line with our expectations. Indeed, it is somewhat surprising that the differential in the reduction in mortality between those who were uninsured at the time of the ACS and the overall population of low-income older adults is not even larger, particularly because they experienced larger gains in Medicaid enrollment;³⁴ however, this may be due to the lower mortality rate in this group.

VIII Interpreting the Estimates and Comparisons to Past Work

VIII.A *Scaling the Mortality Effects by Coverage Changes*

Our results show consistent evidence of a decrease in all-cause mortality among low socioeconomic status, older adults under the ACA Medicaid expansions. In the difference-in-differences model, we estimate an average decrease in annual mortality of 0.132 percentage points during the four-year post period (top panel of Table I). We can combine this estimate with the first stage estimates to provide information on the treatment effect of Medicaid coverage on the group that actually enrolled. For such a scaling to be interpretable as the treatment effect, we must assume that the Medicaid expansions only affected individuals who enrolled in Medicaid. This assumption may be violated if the Medicaid expansions improved access to care more broadly, for example by reducing hospital closures, or improved the health of non-enrollees through other means such as increasing available resources to families that had some uninsured members prior to the expansion.

Our analysis of administrative Medicaid enrollment records indicates that our sample accumulated 0.375 additional years of Medicaid enrollment, on average, as a result of the ACA expansions (see Section V.A). Combining this estimate with the 0.132 percentage point reduction in mortality implies that one year of Medicaid enrollment decreases mortality by about 0.35 percentage points ($= \frac{0.132}{0.375}$). We also find that the probability of enrollment in Medicaid in any given year increased by 12.8 percentage points as a result of the ACA expansions. Assuming that mortality responds only to coverage in a given year, rather than the total amount of coverage experienced, this estimate suggests that contemporaneous enrollment in Medicaid reduces mortality by 1.03 percentage points ($= \frac{0.132}{0.128}$).

To better understand the magnitude of these changes, we would ideally compare them to the mortality rate for sample members who enrolled in Medicaid as a result of the expansions (i.e., the “compliers,” see [Imbens and Angrist, 1994](#)) in the absence of the policy. As discussed earlier, we estimate a counterfactual mortality rate of approximately 1.63 percent for all sample members in the expansion states. However, we expect that the counterfactual mortality rate among the compliers is likely much higher if, for example, those in worse health are more likely to enroll in Medicaid. The literature indicates that such adverse selection does occur (e.g. [Kenney et al., 2012](#); [Marton and Yelowitz, 2015](#)); this may also be exacerbated by policies designed to provide immediate coverage to

³⁴Note that we are unable to estimate the change in Medicaid eligibility or contemporaneous insurance coverage for this subgroup as we did for the other subgroups, due to the lack of information on historical insurance coverage in the ACS. However, we have explored changes in contemporaneous and cumulative insurance coverage for a similar panel in the HRS (see Appendix Table A11). Due to the small sample size, the estimates are imprecisely estimated but do suggest changes in both contemporaneous and cumulative insurance coverage that are larger in size than those for our main sample.